Making a journey

1 8.2 kg = 8200 g. (Fill in the blank)

2 Which rearrangements of speed = distance ÷ time are correct? (Tick **2** correct answers)

- ☐ time = distance × speed
- ☑ time = distance ÷ speed
- ☐ time = speed ÷ distance
- ☑ distance = speed × time
- ☐ distance = speed ÷ time

3 Find the value of x.

distance (miles)	1	5	y
CO_2 emissions (g)	210	x	1680

1050

4 Find the value of y.

distance (miles)	1	5	y
CO_2 emissions (g)	210	x	1680

8

5 A person's car emits 8 kg of carbon dioxide per gallon of fuel and has a fuel efficiency of 50 mpg. Therefore, it produces 160 g of emissions per mile. (Fill in the blank)

$$\text{emissions per mile (g)} = \frac{CO_2 \text{ per gallon of fuel (g)}}{\text{fuel efficiency of the vehicle (mpg)}}$$

6 Aisha walks with an average speed of 5 km/h. It would take 36 minutes to walk 3 km. (Fill in the blank)